

Community Library Management System

A small command-line application for managing a community library: adding books, registering members, lending and returning books, and viewing reports such as overdue loans. It was written in Python to demonstrate core programming concepts — classes, modules, control structures, file handling and exception handling.

Purpose

The system helps a librarian keep track of three things:

- the **catalogue** of books and how many copies of each are available,
- the **members** who are allowed to borrow, and
- the **loans** that link a member to a book, with a due date.

All data is stored in plain CSV files so it persists between runs and is easy to inspect.

Installation and Execution

Requires **Python 3.8+** (uses only the standard library — no packages to install).

```
# 1. Clone the repository
git clone <your-repository-url>
cd library_system
```

```
# 2. Run the program
python main.py
```

On start-up the program loads the CSV files in the `data/` directory. When you choose **Save and exit**, any changes are written back.

Example Usage

```
==== Community Library ====
1. Add book
2. Register member
3. Borrow book
4. Return book
```

5. List books
6. List members
7. Show overdue loans
8. Save and exit

Choose an option (1-8): 3

Member ID: M002

Book ISBN: 9780140328721

Done. Due back on 2026-07-09.

If a rule is broken, the program explains why instead of crashing:

Choose an option (1-8): 3

Member ID: M002

Book ISBN: 9780747532699

Could not lend book: No copies of 'Harry Potter and the Philosopher's Stone' are

Key Features

- Add books (merging copies when an ISBN already exists)
- Register members, each limited to 3 simultaneous loans
- Borrow / return books with automatic due-date tracking (14-day period)
- Overdue-loan report
- Automatic CSV save/load — data survives between sessions
- Clear, custom error messages for every rule violation

Project Structure

File / folder	Responsibility
main.py	Menu-driven user interface (program entry point)
library.py	Library class: rules + CSV save/load
models/book.py	Book class
models/member.py	Member class
models/loan.py	Loan class

<code>exceptions.py</code>	Custom exception types
<code>data/*.csv</code>	Sample data (books, members, loans)